Available online at www.sciencedirect.com

ScienceDirect

IFAC PapersOnLine 59-30 (2025) 353–358

Estimating Human-Exoskeleton Interaction Forces Using an Instrumented Thigh Brace and OpenSim ⋆

Vaibhavsingh Varma * **Zachary Roberts** * **Fawaz Mallick** ** **Mitja Trkov** *

* *Department of Mechanical Engineering, Rowan University, Glassboro 08028 USA. (email: varmav83@rowan.edu, robert75@students.rowan.edu, trkov@rowan.edu)*
** *Department of Biomedical Engineering, Rowan University, Glassboro 08028 USA. (email: mallic26@students.rowan.edu)*

Abstract: Estimating physical human-exoskeleton interaction forces is crucial for developing human-inspired control strategies in assistive devices. To improve our understanding of these interactions, in this paper, we present the design and integration of custom sensor modules to measure these forces and use a simulation approach to estimate them. The developed modules have force-sensitive resistors (FSRs) embedded in custom-designed casings attached to a compliant fabric thigh brace. The instrumented brace is integrated with an existing exoskeleton prototype to measure interaction loads between the user and the device. A dedicated calibration protocol and stationary human-in-the-loop tests were performed to validate the sensor response under various loading conditions mimicking exoskeleton assistance. Subsequently, the instrumented brace was used to record the interaction forces during a walking task performed by three subjects. The results highlight a correlation between interaction loads and exerted load or hip joint motions, as well as inter-subject variability due to large variations in thigh sizes that are an important ergonomic consideration. Finally, the brace was modeled in OpenSim using contact elements and simulated experimental conditions. The simulation results of the interaction forces at a selected contact point were validated by the experimental data.

Copyright © 2025 The Authors. This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0/)

Keywords: Exoskeletons, Force sensitive resistors (FSRs), Instrumented thigh brace, Physical human-machine interactions, Contact simulations, OpenSim

1. INTRODUCTION

Physical human-exoskeleton interactions (pHEIs) play a critical role in the development of exoskeleton technology and affect the comfort of the user and the adoption of technology (Elprama et al., 2024). Measurement and estimation of interaction forces are essential for improving exoskeleton control strategies, enhancing stability, and ensuring user safety. However, most existing lower limb exoskeletons prioritize assistance over precise interaction force sensing, limiting their adaptability and responsiveness (Massardi et al., 2022). Some early exoskeleton prototypes employed direct force measurements at the interfaces (Zoss et al., 2005) or foot-ground contact (Esquenazi et al., 2012; Suzuki et al., 2007) to infer user intent and guide assistance. While these methods have demonstrated the potential for enhancing control algorithms, they are often limited by the complexity and invasiveness of sensor integration.

Recent works on pHEI have increasingly focused on quantifying interaction forces through direct measurements at human-exoskeleton interfaces using force-sensitive resistors (FSRs), multiaxis load cells, and capacitive sensors (Akiyama et al., 2015; Bessler et al., 2019; Choi et al., 2018; Ghonasgi et al., 2021; Serrancolí et al., 2019; Wang et al., 2023). For example, Choi et al. (2018) introduced a simple thigh brace with two FSRs placed along a beam-line structure, using beam deflection to estimate interaction forces. Similarly, Bessler et al. (2019) and Ghonasgi et al. (2021) developed FSR-based interaction force measurement systems for upper limb exoskeletons, employing multisensor arrays to capture distributed forces. Akiyama et al. (2015) used 3-axis force sensors at the exoskeleton-cuff interface to estimate pHEI forces during sit-to-stand and stand-to-sit movements, incorporating a spring-damper model to capture force variations. Furthermore, Serrancolí et al. (2019) employed capacitive sensors combined with a stiffness-damping-based model to characterize pHEI forces during exoskeleton-assisted walking. All of these studies used rigid interfaces or cuffs instead of a soft, compliant brace. Rigid cuffs pose challenges such as pressure concentration and movement discomfort, but have small deformations enabling fast and directional load transfer. On the contrary, compliant braces offer better fit and load distribution, however, their large deformations pose challenges in sensor placements with potential loss in load transfer.

Alongside direct measurement techniques, simulation-based approaches have also emerged as a beneficial tool for estimating pHEI forces and optimizing exoskeleton control strategies. Mosconi et al. (2020) demonstrated the effectiveness of human-exoskeleton interaction modeling in OpenSim (Seth et al., 2011) for testing impedance control. Different studies have proposed spring-like attachments to musculoskeletal models to estimate pHEI forces; however, without experimental validation (De Carvalho et al., 2025; Pang et al., 2022; Yao et al., 2024).

⋆ This material is based upon work supported by the National Science Foundation under Grant No. 2301816. (*Corresponding author: V. Varma*).

2405-8963 Copyright © 2025 The Authors. This is an open access article under the CC BY-NC-ND license.
Peer review under responsibility of International Federation of Automatic Control.
10.1016/j.ifacol.2025.12.262

While these models offer valuable insights, combining experimental force measurements with simulation could enhance the applicability of these models.

In this work, we present the design and characterization of an instrumented thigh brace equipped with custom FSR modules and inertial measurement units (IMUs) to measure pHEI forces and capture movement kinematics. The brace is designed to capture distributed forces at the human-exoskeleton interface during various movements performed by the subjects while wearing a custom exoskeleton prototype developed previously in Varma et al. (2024). The results are demonstrated for a walking task for three subjects with different thigh sizes. In addition, we developed a preliminary thigh-exoskeleton model in OpenSim to mimic pHEI and validated it using the experimental data. Comparisons with other similar studies and limitations of this work are also provided.

The contributions of this work are threefold. First, the developed custom FSR modules and their calibration procedure are novel as they enable measurement of an extended range of pHEI forces by using FSRs with a limited sensitivity range. In addition, the modules are integrated into a soft textile brace to increase user's comfort. Second, the experimental results demonstrate that our instrumented brace with FSR modules can capture interaction forces during walking experiments. Lastly, the developed thigh-exoskeleton OpenSim model can capture the interaction forces and can serve as a tool for further studies enabling better understanding of pHEIs.

2. MATERIALS AND METHODS

2.1 Instrumented Thigh Brace Assembly

The instrumented thigh brace developed in this study is used to measure interaction forces between the previously developed hip exoskeleton reported in Varma et al. (2024) and the wearer's thigh (Fig. 1(a)). The brace employs 8 FSRs placed around the circumference of the thigh in two rows; see Fig. 1(b) and 1(c). The sensor modules were integrated on a compliant fabric brace, which was purposely selected to increase the users' comfort as important design consideration (Elprama et al., 2024). Proper selection and positioning of FSRs are important factors for accurate measurements. Swanson et al. (2019) suggested using FSRs measurements within an expected range of error and calibrating the sensors when measuring interface pressures in lower limb prosthetics. The study suggested that Interlink FSRs exhibit more repeatability, even with drift, and could be used for controlled static and cyclic loading applications with proper calibration, while being successfully used in conditions similar to our application. Therefore, for our study, Interlink FSR-406 sensors were selected due to their relatively large active area (43.69 mm × 43.69 mm), appropriate force measurement resolution/range (0.1N/10N), and cost-effectiveness. In addition, measurement inconsistencies may occur when interfacing with the curved and compliant surface of a human thigh. To mitigate these effects, including preventing the FSRs from bending and ensuring even contact over the FSR, a custom 3D-printed casing was heuristically designed to house the FSRs; see Fig. 2.

The structure of the casing and its dimensional tolerances ensure uniform contact with the complete surface of the FSR via a central pad against the FSR seat (see Fig. 2(c)). The maximum travel of the pad under the load to establish contact with the FSR is 0.2 mm. Importantly, the casing was purposely

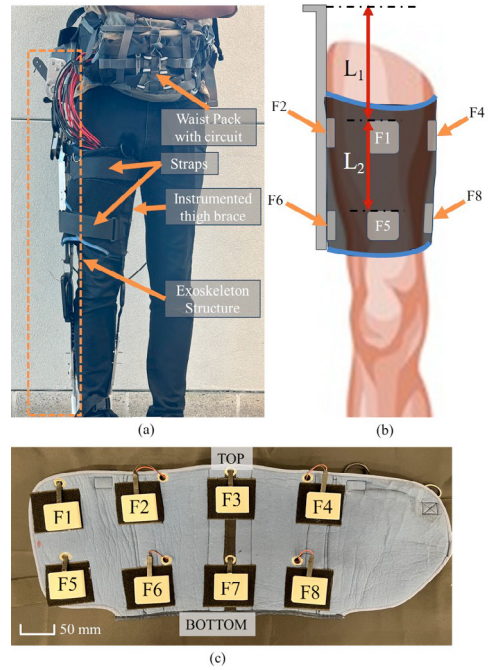

Fig. 1. (a) Subject wearing the instrumented thigh brace with the exoskeleton, (b) Schematic representation of thigh brace and FSR placement in worn condition (exoskeleton link is connected on the lateral side), (c) Instrumented fabric brace with FSR arrangement.

designed to take up some of the initial loading that may be caused by the initial tightening of the brace when a subject dons the brace. Tightening of the brace causes initial loads on the sensor modules that would reduce the sensing range. Offsetting the initial load and introducing an additional stiffness of the casing are both considered advantageous as the functional range of measured loads increases. Wang et al. (2023) reported forces at the shin due to tightening of straps to be within a 10 to 60 N range in the normal standing position, whereas Tamez-Duque et al. (2015) mentioned pressures up to 1000 mm of Hg. Considering these findings, our requirements for the casing were to offset the load for not more than 10 N during the initial tightening. Figures 2 (a-c) show the details of the casing 3D printed out of polylactic acid. The FSRs, positioned within custom housing, required dedicated calibration and human-in-the-loop validation during stationary testing (i.e., tugging while standing) before dynamic testing, such as walking, could be performed. This dynamic testing was conducted in this study and is discussed in the following subsections.

Sensor module placement: FSRs exclusively measure normal pressure components Bessler et al. (2019), and the casing prevents loading due to shear or other means. Therefore, the modules were positioned beneath the two adjustable tightening straps of the brace (Fig. 1). This placement was used to allow contact with the thigh in four directions: anterior, posterior, lateral, and medial (Fig. 1(b)). This simplified configuration can be considered to capture interaction loads around the periphery of the thigh during free movements and device assistance. To allow small variations in placement of module locations to accommodate and properly fit the brace for subjects with variable thigh sizes, the FSRs were mounted using hook and loop straps; see Fig. 1(c). A total of eight modules were integrated into the brace, with four distributed evenly along the upper strap and four along the lower strap.

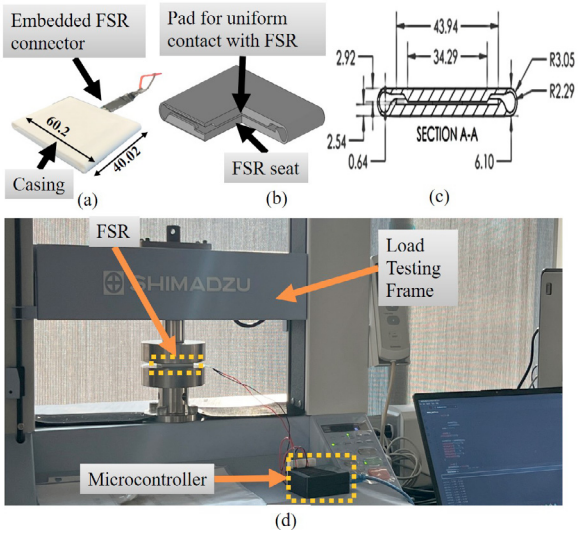

Fig. 2. (a) FSR within casing. (b) Sectional view of the internal geometry of the casing. (c) Dimensional details of casing. (d) Calibration Setup.

Calibration of Sensor modules: The existing literature suggests to perform calibration of the FSRs as used in application condition (Swanson et al., 2019) and employing a non-linear calibration of the FSRs (Ghonasgi et al., 2021). Therefore, we performed a custom FSR calibration while embedded inside the casings by using a universal testing machine (Shimadzu AGS-X); see Fig. 2(d). Compressive loads were applied while recording the sensor voltage output. As recommended by the manufacturer, a voltage divider circuit incorporating a 4.7 kΩ resistor was implemented to optimize the sensor's response range. Measurements from an electromechanical load frame, with a 0.01 mm/min crosshead displacement rate, were recorded simultaneously with the corresponding analog voltage readings obtained via an Arduino Uno microcontroller (see Fig. 2(d)). As the FSRs are meant to be used within the casing, each module was placed within the testing machine and loaded until the FSR saturation was achieved (at around 30 N of applied load). This ensured that the deflection of the casing and its stiffness were inherently taken into account within the calibration model.

2.2 Human-in-the-loop Stationary Testing

In order to evaluate the instrumented brace in the worn condition, stationary human-in-the-loop tests were performed first. Three young, healthy male volunteers performed the tests. The instrumented thigh brace, along with the full exoskeleton equipment, was worn by the subjects. A calibrated S-load cell (FUTEK Advanced Sensor Technology Inc., Irvine, CA) was connected to the exoskeleton thigh link at its center with an inextensible cord and rigidly mounted on the other side. The test setup presented in Fig. 3 shows schematics of subjects applying a load against a fixed instrumented brace and device to a comfortable maximum limit for four testing conditions, namely hip flexion, extension, abduction, and adduction. The subjects were instructed to apply a continuous force for at least 15 seconds. These testing conditions mimic the corresponding applied flexion/extension and abduction/adduction torques provided by the device when actuated. The applied forces and the sensor loads of the resulting interaction forces were recorded using an Arduino MEGA 2560 microcontroller at 150 Hz. The average FSR measurements were compared to the load cell forces to evaluate the nature of the measurements and the most relevant FSR locations for each direction. A Mutual Information test metric (Kraskov et al., 2004) was computed to check which FSRs change mutually with the load cell values.

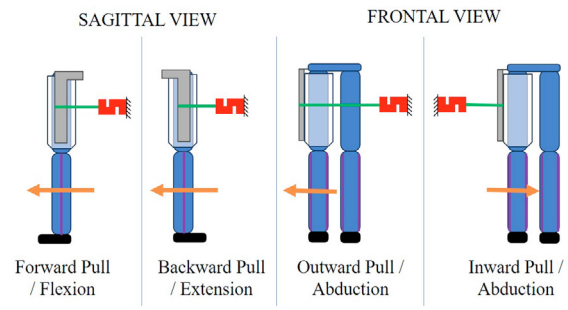

Fig. 3. Human subject for human-in-loop stationary tests setup

2.3 Human Subject Testing During Dynamic Task With Unactuated Exoskeleton

The information about the most relevant sensor readings during tasks from the stationary human-in-the-loop testing was used to monitor those specific FSRs during the dynamic task of walking while wearing the exoskeleton, performed by all three subjects. The purpose was to monitor the interaction forces while wearing the exoskeleton without actuation. This task is of interest for exoskeletons that only provide assistance when needed, such as devices for slip- or trip-induced fall prevention, but otherwise remain in an unactuated mode to save energy.

Before data collection, participants donned the thigh brace and adjusted all straps to a comfortable tightness. Two minutes of adaptation time was provided to the subjects before testing, where the subjects walked around, adjusted, and settled the exoskeleton to best fit their body. The adjustment and adaptation periods also served as a sensor warmup time to minimize the effect of temperature on FSR readings. Three trials were performed for walking by each subject. The data for all trials was post-processed for analysis by aligning and averaging the data for each subject. The main purpose of these tests was to establish a baseline of interaction forces registered by the instrumented brace and to observe inter-subject differences. An Adafruit BNO055 inertial measurement unit (IMU) was also attached to the human subjects' thigh within the brace to measure and record joint angles at 100 Hz while performing the activities. The IMU data was used as the primary kinematic input for preliminary simulations discussed in the next subsection.

2.4 Preliminary Simulation Approach

For this study, a Human-Exoskeleton contact model was developed using OpenSim (Seth et al., 2011), a musculoskeletal modeling and simulation environment. The OpenSim API allows import of custom geometries, joint creation, and inclusion of contact elements that can be added into the model in the form of basic shapes or custom meshes. Figure 4(a) shows a schematic structure of connections between our human musculoskeletal model and the imported exoskeleton model that was adopted from Krüger and Wartzack (2017). A recent study conducted by Yao et al. (2024) presented the development and use of an OpenSim interaction force model in the case of a shoulder support exoskeleton. We follow a similar approach in this study by using the elastic foundation force model (Hast et al., 2019), which is computationally efficient (Halloran et al., 2005), to establish forces between meshes representing the thigh and the

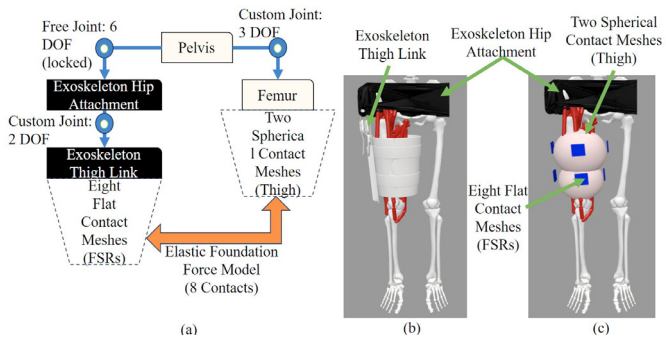

Fig. 4. (a) Virtual model connection tree. (b) Model in simulation Environment. (c) Details of the contact meshes and their positions in the model (exoskeleton thigh link hidden).

sensor modules on the brace. Contact was modeled using virtual stiffness (60,000N/m) and damping (3,000Ns/m), heuristically chosen and gradually tuned to align simulation results with experimental data, though not based on exact human limb properties. Load estimation with the elastic foundation model depends on position, velocity, and area of contact between meshes. Therefore, the thigh was represented by two spherical meshes attached to the right femur, representing the diameter of the thighs measured from the human subjects underneath the top and bottom brace straps. The eight flat meshes attached to the exoskeleton represented the sensor modules; see Fig. 4(b) and 4(c). Simulated conditions were modeled for three subjects, and the initial positions of meshes were determined using actual body measurements (similar to those indicated in Fig. 1(b)).

Finally, quaternion data from the IMU for walking was used to generate hip joint angles using an inverse kinematics algorithm (Stefanov et al., 2020). This formed the basic input for generating controls that drive a Forward Dynamics simulation to estimate contact forces between the thigh and FSR meshes. The estimation is performed within OpenSim using a numerical integration of a general dynamical equation, which includes the contact forces as external loads.

3. RESULTS AND DISCUSSION

3.1 Calibration Model

The average responses of the sensor modules from calibration tests are shown in Fig. 5. The results were used to obtain the relationship between the FSR voltage output (V) and applied load (F). The best fit ($R^2 = 0.9972$) was found with an exponential model: $F = 8.703e^{0.1532V} + 1.703*10^{-5}e^{2.766V}$. The stiffness of the casing affects the sensor load by offsetting the applied load by an average amount of 8.7 N. This is advantageous as the sensor load offsets the initial tightening pressure reading; however, the loads below this threshold cannot be properly detected. Additional advantage of using a casing is that the FSRs effective load range increased from 10 N, as specified by the manufacturer, to 20 N when embedded within the casing. This allows to monitor larger applied loads as the casing material absorbs part of those loads.

3.2 Human-in-the-loop Stationary Testing

The human-in-the-loop testing aimed to ensure that the sensor modules correspond and vary with a known load input. Figure

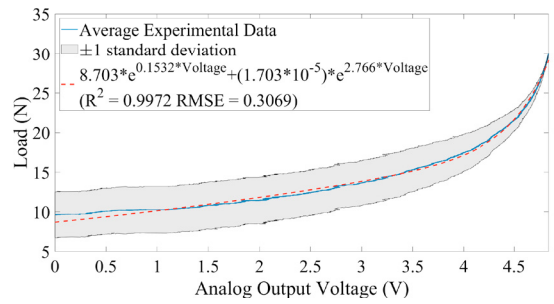

Fig. 5. Calibration curve of the sensor module.

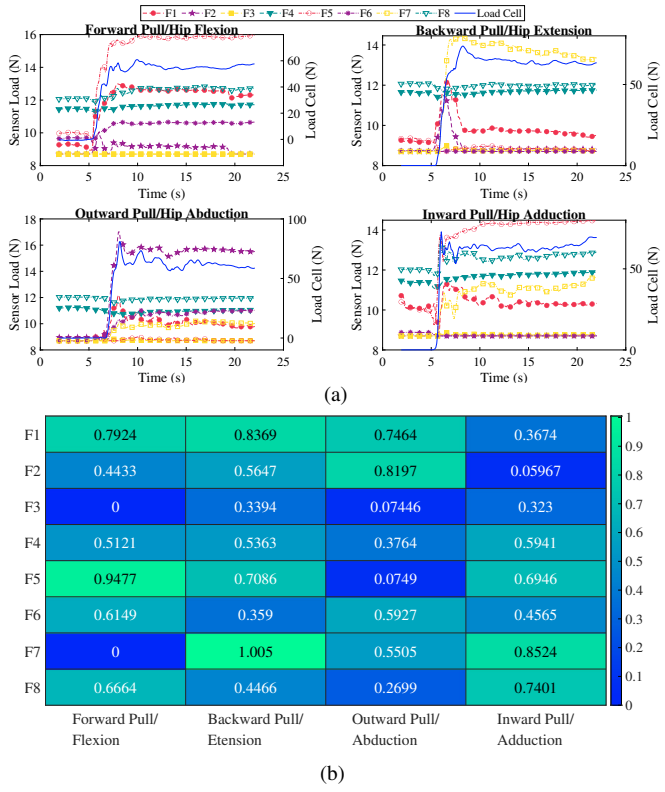

Fig. 6. (a) Load Cell and FSR Response to loading during stationary pulling in different directions. (b) Mutual Information of FSR responses with respect to load cell reading.

6(a) shows the response of all 8 modules to loading in four different directions (see Fig. 3).

The averaged continuous data of FSR measurements is compared to the load cell reading representing the true applied load. In each of the four tests, one or more sensors show a matching trend with the load cell reading, which was further quantified through a statistical metric. Figure 6(b) shows the matrix of mutual information, a statistical metric quantifying the correspondence between the load cell and sensor module readings. An important observation here is that due to the compliant nature of the fabric brace, the applied load is not directly transferred through the modules. The brace compliance and tissue deformation allow the distribution of loads across multiple modules as seen by the metric. The module exhibiting higher mutual information values indicates that it bears a greater portion of the interaction load and can be considered most relevant for a specific loading direction. This insight was used to analyze load variations in the flexion-extension and abduction-adduction directions during the walking task discussed subsequently. We

examined module readings under each strap individually, and selected one FSR from the top (modules F1-F4) and one from the bottom (modules F5-F8) strap alignment, whose reading has the highest correlation. Based on both mutual information metrics and intuitive reasoning, the anterior and posterior modules (F1, F3, F5, and F7) are analyzed in relation to hip flexion-extension, while the medial-lateral modules (F2, F4, F6, and F8) are examined in relation to hip abduction-adduction. Fig. 6(b) shows that in both flexion and extension, module F1 (top strap) provides the highest mutual information with applied load, while F5 and F7 (bottom strap) are most informative. F2 and F4 (top strap) show the highest correspondence for abduction and adduction, respectively, with F6 and F7 (bottom strap) corresponding to the same.

3.3 Interaction Loads for Dynamic Tasks with Unactuated Exoskeleton

The current exoskeleton is designed to allow free movement without actuation, providing assistance only when needed. To support this function, it was important to establish a baseline of interaction loads between the user and the exoskeleton during walking. Fig. 7 presents data from five walking strides across three subjects. It shows FSR load measurements alongside hip joint angles in the flexion-extension and abduction-adduction directions (positive values for flexion and adduction, negative for extension and abduction). For all three subjects, F1 module loading aligns well with hip flexion, while F2 aligns with hip adduction, as shown in Fig. 7. Other modules also track joint angle variations, though with smaller magnitudes. During walking, F1 consistently peaks between 12–16 N across subjects. Some inter-subject differences are observed—for example, Subject 3 shows consistently high F5 readings, while Subject 1 shows lower values. These differences may reflect personal tightness comfort preferences and variations in body size. Subjects 1, 2, and 3 were selected to represent increasing thigh diameters: 143.3 mm, 160.8 mm, and 199.1 mm, respectively.

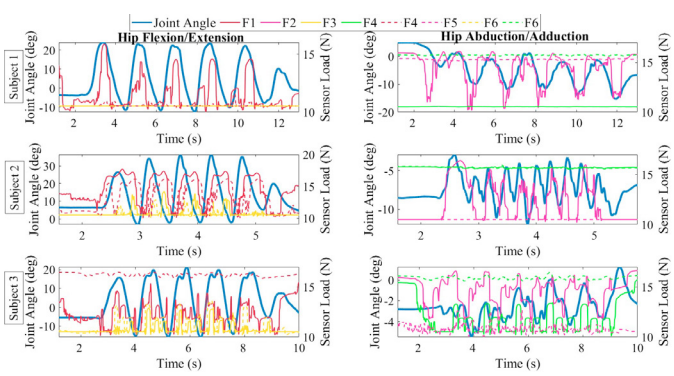

Fig. 7. Interaction loads and joint angle variation for walking task (average over 3 trials each with 5 strides).

While this study presents observations on baseline interaction magnitudes across individuals, the results are limited by noise in the baseline data. Future work can improve accuracy by incorporating filtering techniques and utilizing high-fidelity sensors to better estimate interaction loads. A fully compliant brace introduces uncertainty in load transmission, but the custom casing prohibits shear on the FSR, allowing normal interaction load readings. Load losses are due to fabric compliance, which may not be the case with other rigid braces with inner padding (Ghonasgi et al., 2021; Yao et al., 2024).

3.4 Preliminary OpenSim Simulation Results

We modeled the instrumented brace to estimate contact loads through simulation in OpenSim. The results for a similar approach employed by Yao et al. (2024) show that a simulation model may provide information about the load variation patterns with respect to joint angle change over one elevation cycle; however, the authors also acknowledged the limitation of not achieving similar magnitudes for the simulated and experimental loads. Nevertheless, the results from our simulated model in Fig. 8 show similar timings and load magnitude for the walking test (sensor F1) in multiple gait cycles, which depend on the cyclic nature of the joint angle input for three different subjects.

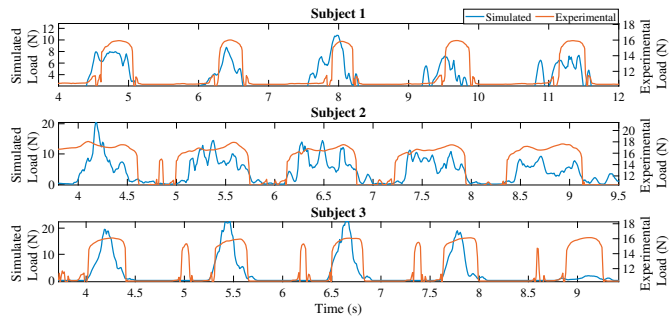

Fig. 8. Comparison of simulated and experimental load profile of sensor module F1 in walking task for three subjects.

Further improvements to the simulated model are required to achieve proper contact on each of the 8 FSR meshes involved. It is also important to note that the simulated environment would need considerable parameter tuning to estimate similar magnitudes as experimental sensor modules for all sensors, which is a challenging task due to soft tissue deformation and muscle contractions. The initial positioning of the simulated sensor modules was also a factor affecting the magnitudes as well as the timing of loads estimated through simulation. Yousaf et al. (2022) have highlighted that the stiffness of human soft tissue changes with angular location, thus, to account for the compliant nature of contact between the sensor modules and the thigh tissue, a contact model with elastic property variation may be necessary.

4. CONCLUSION

In this study, we developed and evaluated an instrumented thigh brace using FSRs housed in custom 3D-printed casings to measure the human-exoskeleton interface forces. A custom calibration established a nonlinear force-voltage relationship, with the casing design expected to help offset the initial tightening loads as well as increase the measured force range. The number of sensors and placement were based on the current exoskeleton thigh strap configuration. Our results suggest that lower sensor density (single row) may suffice for single-strap setups.

Human-in-the-loop testing and dynamic human subject activity trials showed the system could capture directional load variations and inter-subject differences, providing valuable initial baseline of pHEI data. Preliminary simulations reflected similar load timing trends with a promise in matching magnitude

values, but require further refinement for accurate magnitude estimation.

Overall, the brace could be a useful tool for measuring pHEI forces and informing subject-specific exoskeleton control strategies. In the future, further development of the simulation model could enable refined testing in a virtual environment to enable insights in pHEI forces during complex tasks, such as providing external assistance during balance recovery.

REFERENCES

Akiyama, Y., Yamada, Y., and Okamoto, S. (2015). Interaction forces beneath cuffs of physical assistant robots and their motion-based estimation. *Advanced Robotics*, 29(20), 1315–1329.

Bessler, J., Schaake, L., Kelder, R., Buurke, J.H., and Prange-Lasonder, G.B. (2019). Prototype measuring device for assessing interaction forces between human limbs and rehabilitation robots - a proof of concept study. In *2019 IEEE 16th International Conference on Rehabilitation Robotics (ICORR)*, 1109–1114.

Choi, H., Seo, K., Hyung, S., Shim, Y., and Lim, S.C. (2018). Compact hip-force sensor for a gait-assistance exoskeleton system. *Sensors*, 18(2).

De Carvalho, G.B., Chandran, V.D., Spungen, A.M., Bauman, W.A., and Pal, S. (2025). Human-robot dynamics and lower extremity joint forces during exoskeletal-assisted walking in fda-approved rehabilitation robots after spinal cord injury. In J.L. Pons, J. Tornero, and M. Akay (eds.), *Converging Clinical and Engineering Research on Neurorehabilitation V*, 551–554. Springer Nature Switzerland, Cham.

Elprama, S.A., Bock, S.D., Meeusen, R., Pauw, K.D., Vanderborght, B., and Jacobs, A. (2024). Design and implementation requirements for increased acceptance of occupational exoskeletons in an industrial context: A qualitative study. *International Journal of Human–Computer Interaction*, 40(19), 6051–6066.

Esquenazi, A., Talaty, M., Packel, A., and Saulino, M. (2012). The rewalk powered exoskeleton to restore ambulatory function to individuals with thoracic-level motor-complete spinal cord injury. *American Journal of Physical Medicine & Rehabilitation*, 91(11).

Ghonasgi, K., Yousaf, S.N., Esmatloo, P., and Deshpande, A.D. (2021). A modular design for distributed measurement of human–robot interaction forces in wearable devices. *Sensors*, 21(4).

Halloran, J.P., Easley, S.K., Petrella, A.J., and Rullkoetter, P.J. (2005). Comparison of deformable and elastic foundation finite element simulations for predicting knee replacement mechanics. *Journal of Biomechanical Engineering*, 127(5), 813–818.

Hast, M.W., Hanson, B.G., and Baxter, J.R. (2019). Simulating contact using the elastic foundation algorithm in opensim. *Journal of Biomechanics*, 82, 392–396.

Kraskov, A., Stögbauer, H., and Grassberger, P. (2004). Estimating mutual information. *Phys. Rev. E*, 69, 066138.

Krüger, D. and Wartzack, S. (2017). A contact model to simulate human–artifact interaction based on force optimization: implementation and application to the analysis of a training machine. *Computer Methods in Biomechanics and Biomedical Engineering*, 20, 1–10.

Massardi, S., Rodriguez-Cianca, D., Pinto-Fernandez, D., Moreno, J.C., Lancini, M., and Torricelli, D. (2022). Characterization and evaluation of Human-Exoskeleton interaction dynamics: A review. *Sensors (Basel)*, 22(11).

Mosconi, D., Nunes, P.F., Ostan, I., and Siqueira, A.A.G. (2020). Design and validation of a human-exoskeleton model for evaluating interaction controls applied to rehabilitation robotics. In *2020 8th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, 629–634.

Pang, M., Luo, Z., Tang, B., Luo, J., and Xiang, K. (2022). Estimation of the interaction force between human and passive lower limb exoskeleton device during level ground walking. *Biomimetic Intelligence and Robotics*, 2(3), 100056.

Serrancolí, G., Falisse, A., Dembia, C., Vantilt, J., Tanghe, K., Lefeber, D., Jonkers, I., De Schutter, J., and De Groote, F. (2019). Subject-exoskeleton contact model calibration leads to accurate interaction force predictions. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 27(8), 1597–1605.

Seth, A., Sherman, M., Reinbolt, J.A., and Delp, S.L. (2011). Opensim: a musculoskeletal modeling and simulation framework for in silico investigations and exchange. *Procedia IUTAM*, 2, 212–232. IUTAM Symposium on Human Body Dynamics.

Stefanov, K., Huang, B., Li, Z., and Soleymani, M. (2020). Opensense: A platform for multimodal data acquisition and behavior perception. In *Proceedings of the 2020 International Conference on Multimodal Interaction*, ICMI '20, 660–664. Association for Computing Machinery, New York, NY, USA.

Suzuki, K., Mito, G., Kawamoto, H., Hasegawa, Y., and Sankai, Y. (2007). Intention-based walking support for paraplegia patients with robot suit hal. *Advanced Robotics*, 21(12), 1441–1469.

Swanson, E.C., Weathersby, E.J., Cagle, J.C., and Sanders, J.E. (2019). Evaluation of force sensing resistors for the measurement of interface pressures in lower limb prosthetics. *J Biomech Eng*, 141(10), 1010091–10100913.

Tamez-Duque, J., Cobian-Ugalde, R., Kilicarslan, A., Venkatakrishnan, A., Soto, R., and Contreras-Vidal, J.L. (2015). Real-time strap pressure sensor system for powered exoskeletons. *Sensors*, 15(2), 4550–4563.

Varma, V., Patel, S.N., Wilson, N.P., and Trkov, M. (2024). Characterization of hip abduction exoskeleton for assistance during gait perturbations. In *2024 IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*, 70–75. IEEE.

Wang, Y., Qiu, J., Cheng, H., and Zheng, X. (2023). Analysis of human–exoskeleton system interaction for ergonomic design. *Human Factors*, 65(5), 909–922.

Yao, Z., Mir Latifi, S.M., Molz, C., Scherb, D., Löffelmann, C., Sänger, J., Miehling, J., Wartzack, S., Lindenmann, A., Matthiesen, S., and Weidner, R. (2024). A novel approach to simulating realistic exoskeleton behavior in response to human motion. *Robotics*, 13(2).

Yousaf, S.N., Ghonasgi, K., Esmatloo, P., and Deshpande, A.D. (2022). Human-robot interaction: Muscle activation and angular location affect soft tissue stiffness. In *2022 9th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, 01–06. IEEE.

Zoss, A., Kazerooni, H., and Chu, A. (2005). On the mechanical design of the berkeley lower extremity exoskeleton (bleex). In *2005 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 3465–3472.